

12_May-Nov_2020 • Unit 2

1H + 2H source-order past paper

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

Contents

- 1H Q04 • Statistics Toolkit • 4 marks
- 1H Q05 • Powers, Roots & Standard Form • 4 marks
- 1H Q08 • Angles in Polygons & Parallel Lines • 4 marks
- 1H Q09 • Percentages • 3 marks
- 1H Q10 • Prime Factors, HCF & LCM • 3 marks
- 1H Q12 • Simultaneous Equations • 4 marks
- 1H Q13 • Compound Interest & Depreciation • 3 marks
- 1H Q14 • Direct & Inverse Proportion • 3 marks
- 1H Q16 • Circle Theorems • 4 marks
- 1H Q20 • Area & Volume of Similar Shapes • 3 marks
- 1H Q23 • Differentiation • 4 marks
- 1H Q24 • Sequences • 4 marks
- 1H Q25 • Transformations of Graphs • 3 marks
- 2H Q01 • Solving Inequalities • 6 marks
- 2H Q02 • Statistics Toolkit • 5 marks
- 2H Q04 • Ratio Problem Solving • 4 marks
- 2H Q05 • Compound Interest & Depreciation • 6 marks
- 2H Q09 • Volume & Surface Area • 5 marks
- 2H Q10 • Cumulative Frequency Diagrams • 7 marks
- 2H Q11 • Solving Inequalities • 8 marks
- 2H Q15 • Rearranging Formulas • 4 marks
- 2H Q16 • Simultaneous Equations • 5 marks
- 2H Q17 • Solving Inequalities • 5 marks
- 2H Q19 • Vectors • 3 marks
- 2H Q21 • Functions • 7 marks

- 4 Here is a list of six numbers written in order of size.

4 7 x 10 y y

The numbers have

a median of 9

a mean of 11

Find the value of x and the value of y .

$x =$

$y =$

(Total for Question 4 is 4 marks)

- 5 (a) Write 5.7×10^{-3} as an ordinary number.

.....
(1)

- (b) Write 800 000 in standard form.

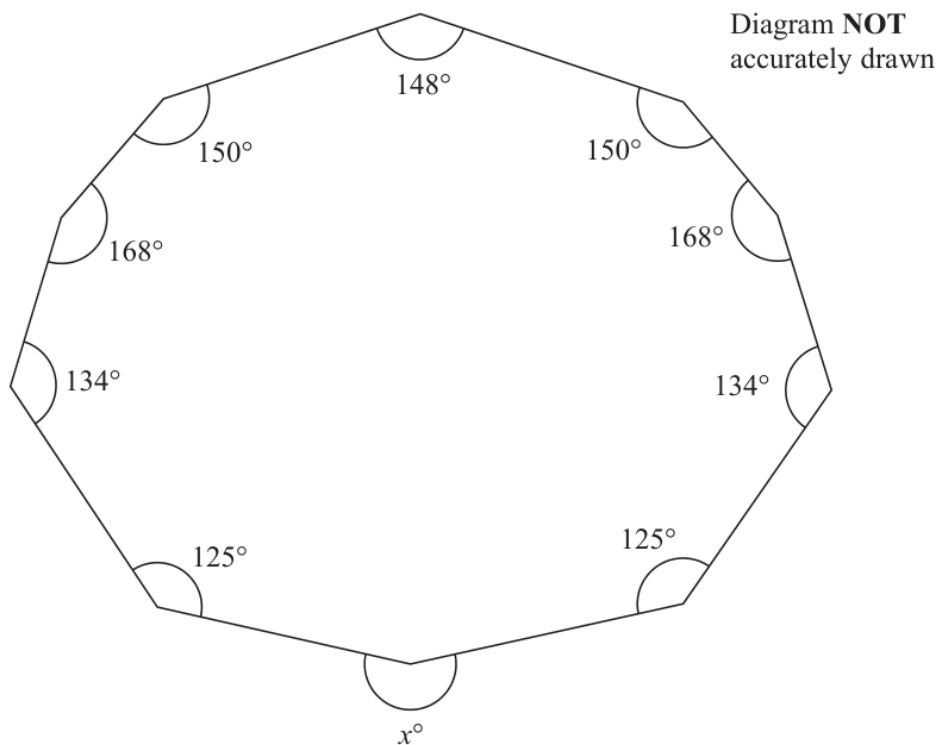
.....
(1)

- (c) Work out $\frac{3 \times 10^5 - 2.7 \times 10^4}{6 \times 10^{-2}}$

.....
(2)

(Total for Question 5 is 4 marks)

- 8 Here is a 10-sided polygon.



Work out the value of x .

$x =$

(Total for Question 8 is 4 marks)

- 9 In a sale, normal prices are reduced by 20%

A bag costs 1080 rupees in the sale.

Work out the normal price of the bag.

..... rupees

(Total for Question 9 is 3 marks)

10 $A = 2 \times 3^{43}$
 $B = 16 \times 3^{37}$

(a) Find the highest common factor (HCF) of A and B .

.....
(1)

(b) Express the number $A \times B$ as a product of powers of its prime factors.
Give your answer in its simplest form.

.....
(2)

.....
(Total for Question 10 is 3 marks)

12 Solve the simultaneous equations

$$\begin{aligned}7x - 2y &= 34 \\ 3x + 5y &= -3\end{aligned}$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 12 is 4 marks)

- 13** Jan invests \$8000 in a savings account.
The account pays compound interest at a rate of $x\%$ per year.
At the end of 6 years, there is a total of \$8877.62 in the account.
Work out the value of x .
Give your answer correct to 2 decimal places.

$x = \dots\dots\dots$

(Total for Question 13 is 3 marks)

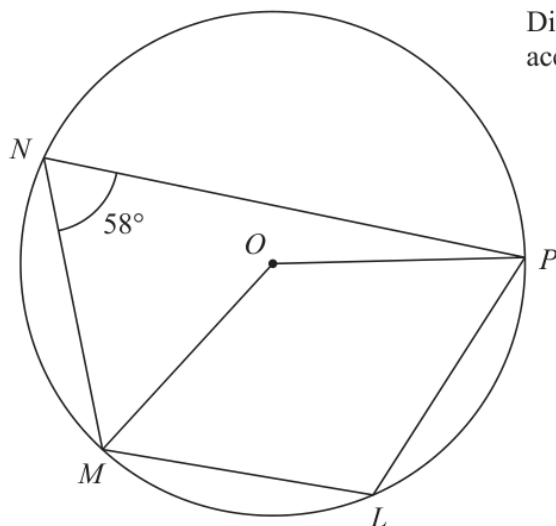
14 F is inversely proportional to the square of v .

Given that $F = 6.5$ when $v = 4$

find a formula for F in terms of v .

(Total for Question 14 is 3 marks)

16

Diagram **NOT**
accurately drawn L, M, N and P are points on a circle, centre O Angle $MNP = 58^\circ$ (a) (i) Find the size of angle MLP

o

(ii) Give a reason for your answer.

(2)

(b) Find the size of the reflex angle MOP

o

(2)

(Total for Question 16 is 4 marks)

20 **R** and **S** are two similar solid shapes.

Shape **R** has surface area 108 cm^2 and volume 135 cm^3

Shape **S** has surface area 300 cm^2

Work out the volume of shape **S**.

..... cm^3

(Total for Question 20 is 3 marks)

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x , in terms of p and m , for which the gradient of **C** is negative.

(Total for Question 23 is 4 marks)

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

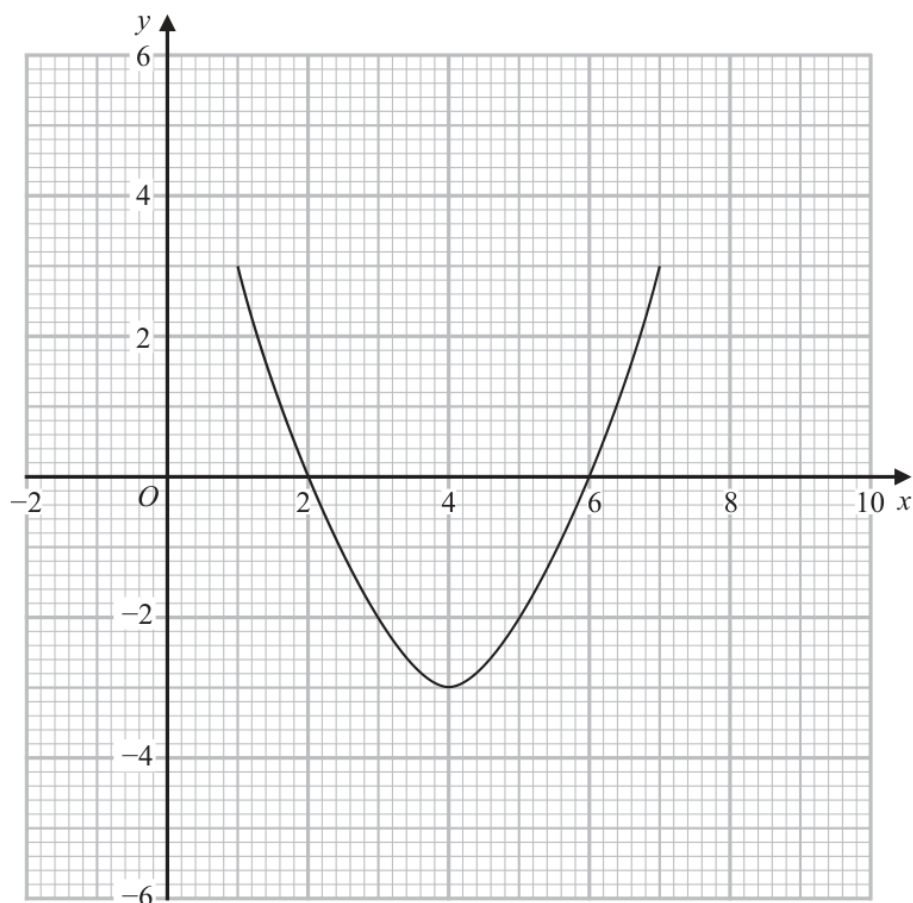
.....
(Total for Question 24 is 4 marks)

- 25 The curve with equation $y = g(x)$ is transformed to the curve with equation $y = -g(x)$ by the single transformation **T**.

(a) Describe fully the transformation **T**.

(1)

The diagram shows the graph of $y = f(x)$



(b) On the grid, draw the graph of $y = 2f(x - 1)$

(2)

(Total for Question 25 is 3 marks)

1 (a) Simplify $g^6 \times g^4$

(1)

(b) Simplify $k^{10} \div k^3$

(1)

(c) Simplify $(3cd^4)^2$

(2)

(d) Solve the inequality $4x + 7 > 2$

(2)

(Total for Question 1 is 6 marks)

- 2 The table shows information about the lengths of time, in minutes, 120 customers spent in a supermarket.

Length of time (L minutes)	Frequency
$20 < L \leq 30$	6
$30 < L \leq 40$	26
$40 < L \leq 50$	31
$50 < L \leq 60$	40
$60 < L \leq 70$	17

- (a) Write down the modal class.

.....
(1)

- (b) Work out an estimate for the mean length of time spent by the 120 customers in the supermarket.

.....minutes
(4)

(Total for Question 2 is 5 marks)

- 4 Andreas, Isla and Paulo share some money in the ratios 3 : 2 : 5

The **total** amount of money that Isla and Paulo receive is £76 more than the amount of money that Andreas receives.

Andreas buys a video game for £48.50 with some of his share of the money.

Work out how much money Andreas has left from his share of the money when he has bought the video game.

£.....

(Total for Question 4 is 4 marks)

- 5 Himari's annual salary is 3 130 000 Japanese Yen (JPY).
She gets a salary increase of 4%

(a) Work out Himari's salary after this increase.

.....JPY
(3)

Kaito bought a car.

The value of the car when Kaito bought it was 750 000 JPY.

At the end of each year, the value of his car had depreciated by 15%

- (b) Work out the value of Kaito's car at the end of 3 years.
Give your answer correct to the nearest JPY.

.....JPY
(3)

(Total for Question 5 is 6 marks)

- 9 The diagram shows a solid cylinder with radius 3 m.

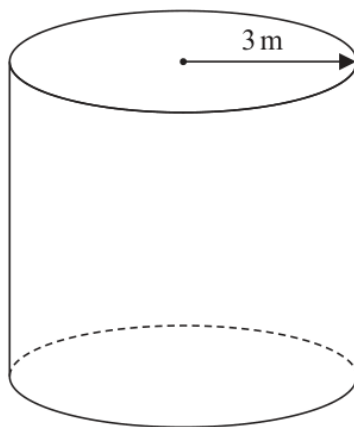


Diagram **NOT**
accurately drawn

The volume of the cylinder is $72\pi \text{ m}^3$

Calculate the **total** surface area of the cylinder.
Give your answer correct to 3 significant figures.

..... m^2

(Total for Question 9 is 5 marks)

10 The table shows information about the number of minutes each of 120 buses was late last Monday.

Number of minutes late (L)	Frequency
$0 < L \leq 10$	10
$10 < L \leq 20$	16
$20 < L \leq 30$	44
$30 < L \leq 40$	29
$40 < L \leq 50$	15
$50 < L \leq 60$	6

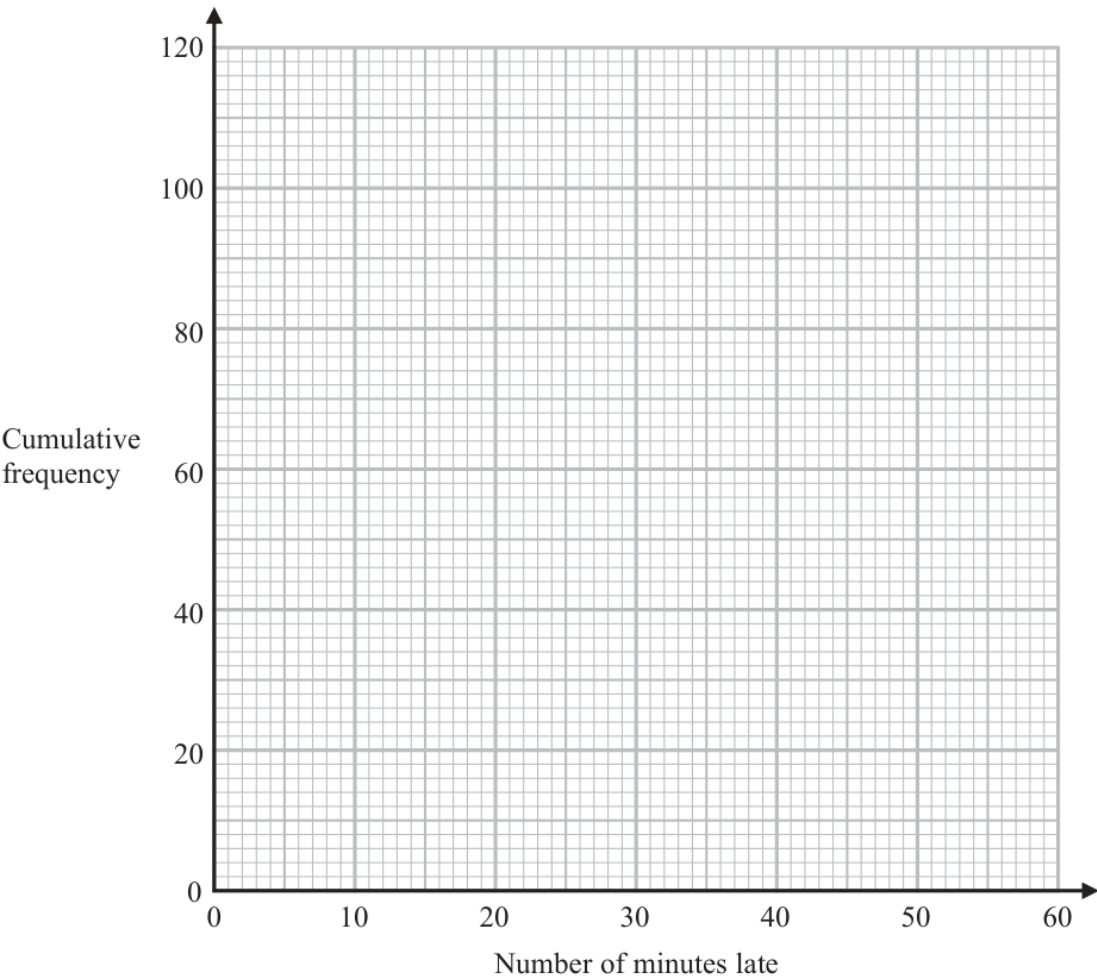
(a) Complete the cumulative frequency table below.

Number of minutes late (L)	Cumulative frequency
$0 < L \leq 10$	
$0 < L \leq 20$	
$0 < L \leq 30$	
$0 < L \leq 40$	
$0 < L \leq 50$	
$0 < L \leq 60$	

(1)



(b) On the grid, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the interquartile range.

.....minutes
(2)

(d) Use your graph to find an estimate for the number of buses that were more than 48 minutes late last Monday.

.....
(2)

(Total for Question 10 is 7 marks)

11 (a) Simplify fully $(8e^{15})^{\frac{2}{3}}$

(2)

(b) Express $\left(\frac{y}{2}\right)^{-4}$ in the form ay^n where a and n are integers.

(2)

(c) Solve $\frac{4x-2}{3} - \frac{5-3x}{4} = 6$

Show clear algebraic working.

$x =$
(4)

(Total for Question 11 is 8 marks)

15 Make x the subject of $y = \frac{5 - 2x}{x + 3}$

(Total for Question 15 is 4 marks)

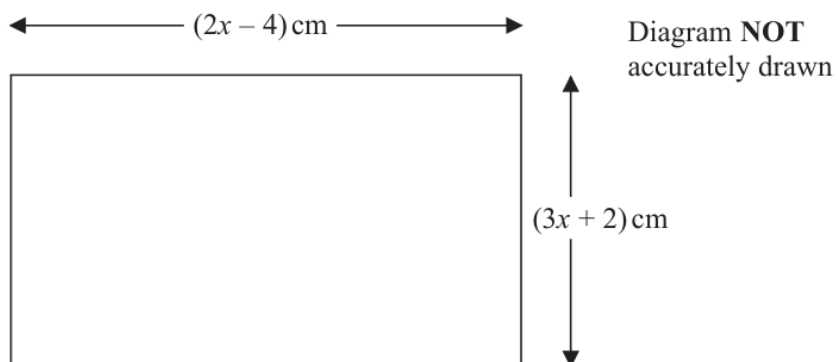
16 Solve the simultaneous equations

$$\begin{aligned} 3xy - y^2 &= 8 \\ x - 2y &= 1 \end{aligned}$$

Show clear algebraic working.

(Total for Question 16 is 5 marks)

17 The diagram shows a rectangle.



The area of the rectangle is $A \text{ cm}^2$

Given that $A < 3x + 27$

find the range of possible values for x .

(Total for Question 17 is 5 marks)

19 OAB is a triangle.

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

The point C lies on OA such that $OC : CA = 1 : 2$

The point D lies on OB such that $OD : DB = 1 : 2$

Using a vector method, prove that $ABDC$ is a trapezium.

(Total for Question 19 is 3 marks)

21 The function f is such that $f(x) = 5 + 6x - x^2$ for $x \leq 3$

(a) Express $5 + 6x - x^2$ in the form $p - (x - q)^2$ where p and q are constants.

.....
(2)

(b) Using your answer to part (a), find the range of values of x for which $f^{-1}(x)$ is positive.

.....
(5)

(Total for Question 21 is 7 marks)